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ORIGINAL INVESTIGATION

The Effect of a Synthetic-Grass Sport Surface on Physiology and Perception During Intermittent Exercise in Hot Conditions

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Purpose: The current study aimed to determine the effect of a synthetic-grass sport surface on core body temperature, skin temperature, heart rate, thermal sensation, thermal comfort, and rating of perceived exertion (RPE) during intermittent exercise in hot conditions. ***Methods:*** Using a randomized crossover design, 13 trained/developmental team-sport athletes completed two 50-minute standardized intermittent running protocols on a synthetic and a natural-grass surface, on separate days (control-condition air temperature 32.6 °C [1.3 °C], relative humidity 43.2% [5.3%]). ***Results:*** Final skin temperature was significantly higher on synthetic compared with natural grass at the calf (40.1 °C [2.5 °C] vs 33.4 °C [0.6 °C]; $P < .001$), shoulder (36.6 °C [1.7 °C] vs 33.7 °C [0.7 °C]; $P < .001$), and chest (33.2 °C [1.1 °C] vs 31.8 °C [1.2 °C]; $P = .02$). Thermal sensation (median: 2.3; interquartile range [0.5] vs 2.2 [0.5], $P = .03$) and sweat rate (1.5 [0.4] $L \cdot h^{-1}$ vs 1.2 [0.3] $L \cdot h^{-1}$; $P = .02$) were also significantly higher on synthetic grass. While final core body temperature was significantly higher on the natural than synthetic grass (38.4 °C [0.3 °C] vs 38.2 °C [0.4 °C]), there were no significant differences in delta core temperature, as well as heart rate, thermal comfort, or RPE. ***Conclusions:*** Higher skin temperatures, thermal sensation, and sweat rates suggest that exercising on synthetic grass in hot conditions may increase some markers of heat strain during exercise. However, delta core body temperature, heart rate, thermal comfort, and RPE remained unaffected.

Keywords: thermoregulation, heat strain, artificial turf, core body temperature, thermal sensation

Synthetic grass or artificial turf surfaces have become a popular alternative to natural grass in sports due to practical advantages (eg, high usage hours) over natural grass.[1,2] Recently, the introduction of synthetic grass sports surfaces has created controversy, with studies raising concerns regarding high surface temperatures on synthetic grass surfaces influencing the surrounding microclimate and increasing the level of absorbed radiation that an athlete may experience.[3–5] For instance, a recent systematic review suggested that air temperatures are up to 2.8° C higher over synthetic than natural grass[6] and the mean radiant temperature up to 3.4° C higher above synthetic than natural grass.[7] In hot environments with a higher air and mean radiant temperature than an athlete's core, skin, and mean body temperature, the capacity for heat loss is reduced because the skin-air temperature gradient is negative, meaning that an athlete can gain heat from the environment,[8] potentially elevating the risk of heat strain. In addition, athletes competing in the heat can change their behavior to alleviate heat strain, such as reducing the volume of high speed running,[9] or making modifications to their pacing strategies,[10] which can affect performance. Given the importance of these environmental parameters on human thermoregulation, the increased air and mean radiant temperatures over synthetic grass surfaces raise an interesting question: Whether the changes in the surrounding environment can affect human heat exchange and cause meaningful heat strain for athletes during training and competition?

Several studies have investigated whether synthetic grass surfaces can increase heat stress.[5,7,11,12] Statistical modeling has been performed to estimate the energy budget (ie, the sum of heat gain and heat loss).[5,7] These studies estimated that heat stress is higher when exercising on synthetic surfaces compared to natural grass surfaces.[5,7] For instance, one study reported that the calculated heat budget was between 4.68 $W \cdot m^2$ and 26.95 $W \cdot m^2$ higher for adults and children exercising on a synthetic than a natural grass surface.[5] Nevertheless, it is unknown whether these differences are enough to affect physiological markers of heat strain, as these calculations are based on theoretical models. Only one investigation has measured the heat strain risk using physiological data derived from collegiate American Football athletes during a training session, where the athletes experienced a significantly higher mean skin temperature by 1.9 °C and significantly higher perceived heat stress on a synthetic compared to a natural grass field in hot conditions.[12] However, this research was limited as trial order was not randomized, and the training demands were not standardized completely across trials. Therefore, a well-controlled experiment is required to confirm these data.

Given the limited human data, and the increasing popularity of synthetic sports surfaces, a recent Australian government report has called for research that determines whether *synthetic grass surfaces affect markers of heat stress and thermoregulation in humans.*[13] Additionally, an International Olympic Committee consensus statement on recommendations and regulations for sport events in the heat also called for researchers to investigate the heat load experienced by athletes on different sports surfaces.[14] Therefore, the current study aimed to determine the effect of a synthetic grass sports surface on physiology and perception during intermittent exercise in hot conditions. Specifically, the primary outcome measure of interest was the athletes' core body temperature, while secondary measures included skin temperature, heart rate, sweat rate, rating of perceived exertion (RPE), thermal sensation, and thermal comfort during intermittent running in hot conditions. It

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was hypothesized that athletes would experience increased core body temperatures while exercising on the synthetic surface compared to natural grass.

Methods

Subjects

Thirteen trained/developmental[15] team-sport athletes (9 male, 4 female) volunteered for the study (mean [SD]; age: 29 [7.9]; stature: 173.0 [7.7] cm; body mass: 79.0 [13.4] kg; fat %: 18.4% [12.7%]; 30–15 Intermittent Fitness test score: 15.8 [1.8]). The sample size was based on an a priori power analysis for a paired sample *t* test (G*Power, version 3.1.9) using a 1-tailed *t* test with alpha set at .05, power at 0.8, and effect size of 0.8, which determined a total sample size of 12 was required. This study was performed in February 2023, meaning that the athletes were likely partially acclimatized from outdoor exposures during the Australian summer period. Athletes were required to be ≥18 years old, have participated in a competitive team sport within the last 12 months, and pass the Exercise and Sport Science Australia Adult Pre-Exercise Screening System.[16] All procedures for the investigation were approved by the Southern Cross University Human Research Ethics Committee in the spirit of the Declaration of Helsinki (2022/150), and athletes' provided consent prior to engaging in all study procedures.

Design

Athletes attended 3 separate sessions spaced 1 week apart each, including a preliminary testing session and 2 experimental trials. The athletes' cardiorespiratory fitness was assessed during the preliminary testing session. Then, with a randomized and balanced (as best as possible) cross-over experimental design, athletes completed 2 separate 50-minute standardized intermittent running protocols on a synthetic (Ligaturf HB250, Polytan) and natural grass surface on hot days.

Preliminary Testing Session

During the preliminary testing session, stature was measured using a portable stadiometer (Seca 213), and body mass and body fat percentage were recorded using a scale with bioelectric impedance analysis (Tanita BC-587). Athletes' cardiorespiratory fitness was assessed using the "30–15 intermittent fitness test" on a natural grass field. The final running velocity was used to calculate the athletes' maximal aerobic speed (MAS).[17] The 30–15 intermittent fitness test has an excellent test–retest reliability for maximal velocity.[18] Athletes were also familiarized with 3 perceptual scales (ie, thermal sensation, thermal comfort, and RPE) before and after the 30–15 running test, to be used during the experimental trials.

Experimental Trials

All athletes completed each experimental trial on the same 2 days, spaced 1 week apart. During the first trial, the athletes were randomly split into 2 groups using a randomized number generator, with 1 group assigned to the synthetic grass field and the other to the natural grass field. In order to achieve balanced groups, a number generator was used until 1 group was assigned 7 athletes, and then, the remaining were placed in the second group. Experimental running trials were performed at the C.ex International Stadium, Coffs Harbour, Australia, and the 2 fields were approximately 50 m apart. All athletes completed the running protocol simultaneously on each day, enabling the athletes to complete the trial under the same environmental conditions each day. A standardized warm-up was performed on the athletes' respective field (ankle jumps, 20 m jog through, 10 m high knee march, 2 × 10 m high knee skips, 20 m side skips, 20 m carioca, leg swings, 3 × 20 m sprints). The running protocol involved 4 × 6 laps of intermittent running at different percentages of their individual MAS, so that the running demands were standardized within and between athletes. One lap of the protocol involved two 15-second efforts at 100% of their MAS and two 15-second recovery periods at 60% of their MAS. Athletes received a 30-second rest between each lap and a 3-minute rest between each set. All athletes had their running distances marked out on the surface, and a speaker was used to provide an audio signal every 15 seconds to ensure they reached their running distances within the allocated time. Athletes also consumed 600 mL of thermoneutral water during the experimental trials to standardize fluid intake. This volume of water was selected due to the size of the provided water bottles. The athletes were provided a standardized white polyester sports singlet to wear throughout each running trial. Additionally, athletes were also required to wear sports shorts, socks, and football shoes, and were instructed to wear the same clothing during both running trials.

Temperature Measurements

For the measurement of core body temperature, athletes ingested a telemetric capsule (BodyCap) on the morning of each experimental trial approximately 5 hours before baseline measurement, allowing the capsule to pass into the lower intestine.[19,20] Core body temperatures were sampled at 30-second intervals during the running trials and later downloaded with a wireless data receiver (e-Viewer; BodyCap). The e-Celsius system has been used in intermittent sports research[20,21] and has excellent validity and test–retest reliability.[19]

Skin temperatures were measured before and immediately after the running trials using a dermal thermometer (Derma Temp) at 3 anatomical locations, including the left lateral deltoid (shoulder), left pectoralis major (chest), and left gastrocnemius (calf). The average of 2 measurements obtained at each site was taken.

Heart-Rate Measurements

Chest-strapped heart rate monitors (Polar T31) provided heart-rate measurements telemetrically at 5 Hz throughout the running trials. The heart rates were recorded after each set of the running protocol.

Perceptual Measurements

The athletes' RPE, thermal sensation, and thermal comfort were recorded at baseline and after each set of the running protocol. Borg RPE is a valid psychophysical scale (where 6 = none and 20 = very hard).[22] Thermal sensation is a 9-point category ratio (where –4 = very cold and 4 = very hot).[23] Thermal comfort is a 10-point ratio scale (where –4 = very uncomfortable and 4 = very comfortable).[23]

Hydration Status

Athletes were asked to void their bladder and provide a midstream urine sample before the running trials. This sample was then used to perform a urine-specific gravity measurement with a hand-held refractometer (PAL-1BLT/I). If the urine-specific gravity analysis was >1.020, athletes consumed 500 mL of thermoneutral water over a 20-minute period before commencing the protocol.

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Sweat Rate

All athletes were towel-dried and weighed on scales (Tanita MC-780U) in minimal clothing before and after exercise. The difference in body mass (adjusted for fluid intake and any urine excreted) was used to estimate the whole-body sweat loss and rate.[24]

Environmental Conditions

Heat stress meters with data logging capacity (Reed Instruments SD-2010) were placed on each surface, which recorded air temperature, relative humidity, and wet-bulb globe temperature at 30-second intervals throughout the trials. The heat stress meters were positioned 1.2 m above the center of both fields as per recommendations.[14] Surface temperatures were recorded before and immediately after the running trials using a thermal imaging camera (FLIR-E5, Teledyne Technologies) positioned 1 m above the center of each field.

Statistical Analysis

This study used a series of statistical analyses tailored to the nature of each variable using JASP software (JASP Team, version 0.19.0)[25]. Comparisons of overall group means for core body temperature, skin temperature, sweat rate, urine-specific gravity, and environmental parameters were assessed using paired samples *t* tests. These results are reported as means and delta changes, with 95% CIs. Wilcoxon signed-rank tests were used to compare the median differences between thermal sensation, thermal comfort, and RPE, with 95% CIs for Hodges–Lehmann estimates.

Repeated-measures data, including core body temperature, heart rate, and RPE, were analyzed using linear mixed-effect models. These models accounted for the repeated observations within participants and allowed for the assessment of fixed effects (ie, surface type and time point) and their interaction. To accommodate for the repeated-measures design and control for nonindependence of observations within subjects, participants were included as a random effect. Pairwise comparisons for significant interactions were made by creating custom contrasts under the estimated marginal means framework within JASP. Results from the linear mixed effect models were reported as means with 95% CIs. Repeated measures of ordinal data (ie, thermal sensation and comfort) were analyzed across time using the Friedman tests. Statistical significance was set at $P \leq .05$ for all analyses.

Results

The environmental conditions for each trial are presented in Table 1. Environmental conditions over the natural grass surface (ie, the control condition) during days 1 and 2 were similar. For days 1 and 2, the mean air temperature was 33.9 °C (0.5 °C) and 31.4 °C (0.6 °C), the mean relative humidity was 37.7% (2.5%) and 48.0% (1.3%) and the mean wet-bulb-globe temperature was 26.4 °C (0.2 °C) and 27.0 °C (0.3 °C). The mean running distance (synthetic: 4608 [458] m; natural: 4608 [458] m) and running speeds for intervals at 100% MAS (synthetic: 4.4 [0.5] $m \cdot s^{-1}$; natural: 4.4 [0.5] $m \cdot s^{-1}$) and 60% MAS (synthetic: 2.7 [0.3] $m \cdot s^{-1}$: natural: 2.7 [0.3] $m \cdot s^{-1}$) were exactly the same for each surface as they were standardized. There was no trial order effect for any outcome measure.

A summary of the overall physiological and perceptual responses between surfaces across the running protocol is presented in Tables 2 and 3. While there was a significant difference in absolute core body temperature, there were no significant differences for delta core body temperature between surfaces. Absolute and delta changes in shoulder, chest, and calf skin temperatures were significantly higher on synthetic compared with natural grass at the end of the trials. Sweat rate and thermal sensation were also significantly higher on synthetic compared with, natural grass.

The individual responses for delta core body temperature and time course of the core body temperature measurements across the running protocol on the synthetic and natural grass surface are illustrated in Figure 1.

There was no significant interaction effect for core body temperature ($P = .49$), heart rate ($P = .55$), and RPE ($P = .32$) between the surface and time across the protocol. However, there were significant fixed effects across time for core body temperature ($P < .001$), heart rate ($P < .001$), and RPE ($P < .001$), with the pairwise comparisons presented in Supplementary Material (available online). There were significant differences in thermal sensation and thermal comfort over time on synthetic and natural grass (ie, within surfaces; $P < .001$). There was only a significant difference for thermal sensation postset 1 between synthetic and natural grass 1.0 (1.0, 1.0), $P < .001$; see Supplementary Material (available online).

Discussion

This study aimed to determine the effect of a synthetic grass surface on trained/developmental team sport athletes' physiology and perception during intermittent exercise in hot conditions. It was hypothesized that athletes would experience increased core body temperatures while exercising on the synthetic surface. The findings demonstrated that the athletes' delta core body temperature was not significantly different between surfaces nor was heart rate, RPE, and thermal comfort. However, the athletes experienced

Table 1 Summary of the Environmental Parameters Between Surfaces Across the Running Protocol for Each Trial

	Synthetic, mean (SD)	Natural, mean (SD)	Mean difference	95% CI	Significance *P*
Air temperature, °C	35.8 (3.4)*	32.6 (1.3)	3.3	2.0 to 4.6	<.001
Relative humidity, %	35.9 (9.8)*	43.2 (5.3)	−7.4	−10.1 to −4.7	<.001
WBGT, °C	27.4 (0.1)*	26.7 (0.3)	0.7	0.5 to 9.4	.001
Wind velocity, $m \cdot s^{-1}$	2.4 (2.2)	2.4 (2.2)	0.0	—	
Surface temperature, °C	67.4 (0.1)*	37.0 (0.9)	30.4	29.9 to 30.9	<.001

Abbreviation: WBGT, wet-bulb globe temperature.
*Significantly different compared to natural grass.

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Table 2 Summary of the Physiological Responses Between Surfaces Across the Running Protocol for Each Trial

Variable	Synthetic, mean (SD)	Natural, mean (SD)	Mean difference	95% CI	Significance, *P*
Core body temperature, °C	38.2 (0.4)*	38.4 (0.3)	0.2	–0.3 to 0.0	.042
Heart rate, beats·min^{-1}	167.0 (7.0)	169.0 (8.0)	–1.7	–6.4 to 3.0	.44
Sweat rate, L·h^{-1}	1.5 (0.4)*	1.2 (0.3)	0.3	0.1 to 0.5	.02
Urine-specific gravity	1.019 (0.006)	1.022 (0.007)	–0.002	–0.008 to 0.003	.40
Post set 4 chest skin temperature, °C	33.2 (1.1)*	31.8 (1.2)	1.3	0.2 to 2.6	.02
Post set 4 shoulder skin temperature, °C	36.6 (1.7)*	33.7 (0.7)	2.9	1.7 to 4.2	< .001
Post set 4 leg skin temperature, °C	40.1 (2.5)*	33.4 (0.6)	6.6	5.2 to 8.1	< .001
ΔCore body temperature, °C	1.3 (0.3)	1.4 (0.5)	–0.1	–0.4 to 0.2	.50
ΔChest skin temperature, °C	–0.5 (1.4)*	–1.8 (1.5)	1.3	0.0 to 2.7	.05
ΔShoulder skin temperature, °C	2.5 (2.2)*	–0.6 (1.3)	3.1	1.6 to 4.6	< .001
ΔCalf skin temperature, °C	7.1 (2.9)*	0.5 (0.9)	6.6	4.8 to 8.4	< .001

*Significantly different compared to natural grass.

Table 3 Summary of the Perceptual Responses Between Surfaces Across the Running Protocol for Each Trial

	Synthetic median, IQR	Natural median, IQR	Hodges–Lehmann estimate	95% CI	Significance, *P*
RPE, AU	14 (0.8)	15 (0.7)	0.5	–2.0 to 1.0	.60
Thermal comfort, AU	2.5 (0.7)	2.0 (0.8)	0.6	–0.3 to 0.6	.56
Thermal sensation, AU	2.3 (0.5)*	2.2 (0.5)	0.3	0.0 to 0.6	.03

Abbreviations: AU, arbitrary units; 95% CI, confidence interval for the Hodges–Lehmann estimate; IQR, interquartile range; RPE, rating of perceived exertion.
*Significantly different compared with natural.

significantly increased skin temperatures, sweat rate, and thermal sensation on the synthetic surface compared to natural grass.

Human skin temperatures are expected to increase during exercise in the heat due to peripheral blood flow and direct exposure to the thermal environment.[26] In the current study, the air temperature was higher over the synthetic surface, which likely caused higher skin temperatures. A previous study also reported that collegiate American Football athletes experienced a significantly higher average skin temperature during training on synthetic grass (37.1 °C [0.5 °C]) than natural grass (35.2 °C [0.3 °C]).[12] Synthetic grass surfaces emit more longwave radiation than natural grass surfaces,[3,4,12] elevating the air temperature and absorbed radiation directly above the surface. A recent systematic review demonstrated that air temperatures over synthetic grass can exceed natural grass by 0.5 °C to 1.2 °C at heights between 1.2 and 1.5 m above the surface.[6] Interestingly, temperatures increased closer to the surface, with maximum air temperature differences 15 cm above synthetic and natural grass reported between 5.2 °C and 8.4 °C.[3–5] Indeed, this study also observed mean air temperatures being 3.3 °C higher 1.2 m over synthetic compared with natural grass. Given that air temperatures are likely hotter closer to the surface, skin surfaces closer to the synthetic grass (ie, on the calf) would likely have the most significant increases in skin temperature, as observed in the present study, with calf temperatures having the highest increase compared to the other sites. Athletes also exercised in singlets on both surfaces, exposing their shoulders to direct sunlight, and it can be assumed that the athletes on both surfaces experienced the same amount of direct sunlight because the surfaces were positioned adjacently. However, since synthetic grass surfaces increase air temperatures directly above the surface, this may have added more heat to the shoulder and chest skin sites, resulting in a greater change in skin temperature on the synthetic than on the natural grass surface. Additionally, the increased skin temperatures also likely increased the sweat rate on the synthetic grass surface to counteract rapid body temperature changes, especially given that there were no significant differences for urine-specific gravity between surfaces. It should be noted that chest skin temperatures on both surfaces decreased from baseline, which may be explained by sweat cooling the skin surface, and as such, it is likely that the singlet was saturated by the sweat, cooling the skin site.

The athletes in the present study reported a higher thermal sensation on the synthetic grass than on the natural grass surface. Given that skin temperatures were higher on the synthetic grass, it is not surprising that the athletes felt hotter. Autonomic temperature regulation in humans is triggered by a complex network of thermoreceptors in the skin, which detect changes in skin temperature and relay this information to higher order centers within the brain to initiate appropriate responses.[27] When skin or body temperature changes are detected, a thermal sensation is elicited, and the magnitude of this sensation is influenced by the exposed heat load.[27] Since the synthetic grass surface emitted a higher heat load directly above the surface than natural grass, resulting in increased skin temperatures, this likely triggered a higher thermal sensation for athletes exercising on the synthetic grass surface. Similarly to the current study, a recent investigation reported athletes perceived more heat stress while training on a synthetic grass surface (heat scale median: 5.0; first and third quartile: 4.0–

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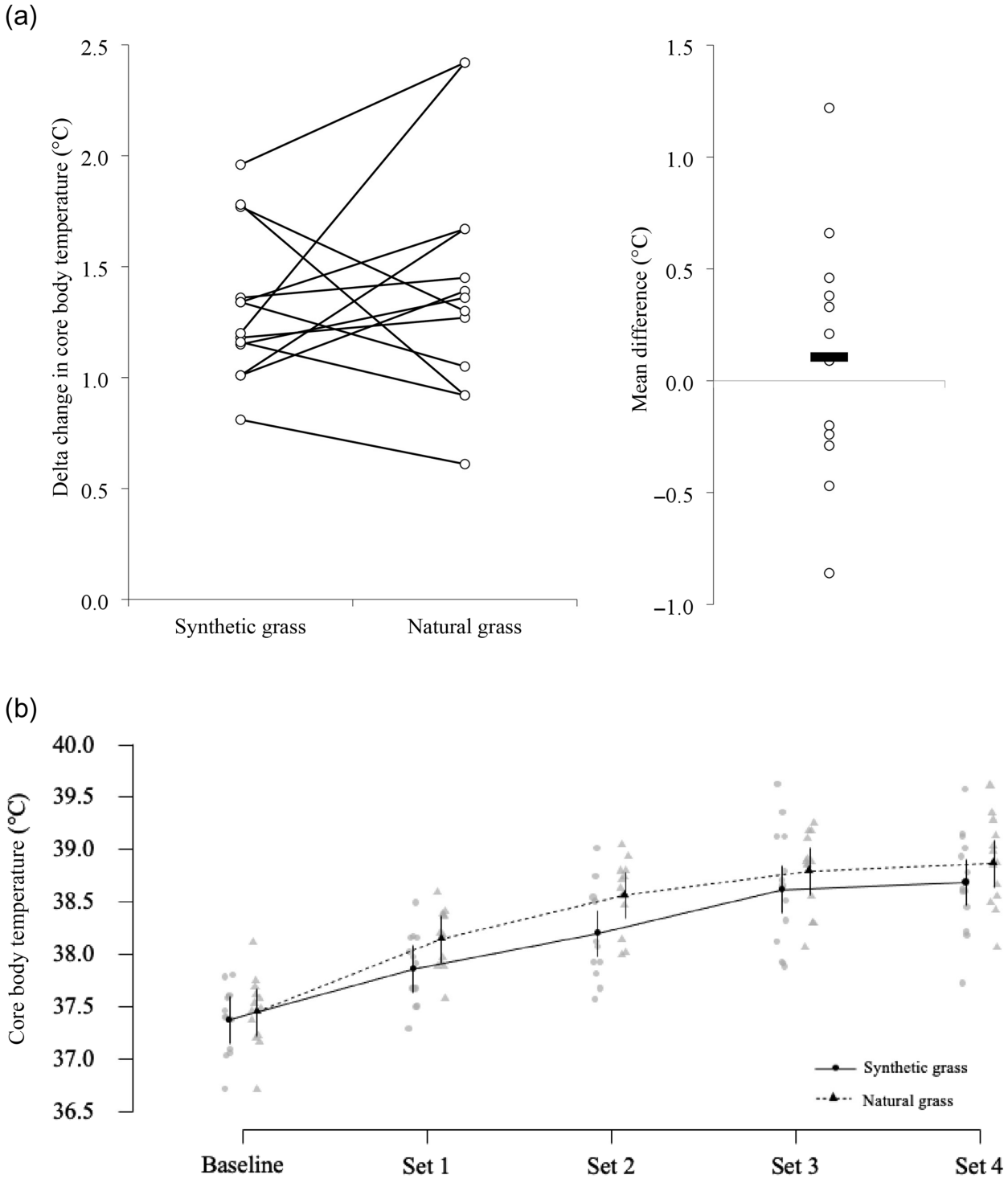

Figure 1 — (a) Individual delta changes in core body temperature from the running protocol. (b) Time course of the core body temperature measurements across the running protocol. *Note*. (a) Individual and mean responses in core body temperature from the running protocol presented as dots and for each surface are illustrated using a horizontal straight line. (b) Individual athlete responses for mean core body temperature at each time point (dots) and 95% CIs for the mean core temperature at each time point (horizontal lines).

6.0) than a natural grass surface (heat scale median: 4.0; first and third quartile: 4.0–5.5) using a 10-point Likert scale.[12] However, athletes in the present study reported a similar thermal comfort on both surfaces, suggesting that the increased thermal sensation might not have caused more discomfort to the exercising athletes. This may be explained by the intermittent nature of the running protocol, involving maximal bouts of exercise followed by recovery periods. The rest periods and regular access to water may have allowed the athletes to recover enough to avoid a meaningful increase in thermal discomfort.

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While athletes experienced higher changes in skin temperatures and thermal sensation, differences were not observed for core body temperature, heart rate, and RPE between the 2 surfaces. Indeed, previous research has suggested that synthetic grass surfaces could cause more human-absorbed longwave radiation than natural grass and increase human heat gain.[3–5] However, in line with our findings, a separate study that compared heat strain markers on a synthetic and natural grass surface did not observe any significant differences in athletes core body temperatures and RPE.[12] For example, the aforementioned study on collegiate American Football athletes reported no significant change in core body temperatures and RPE during training sessions on a synthetic (core body temperature: 37.6 °C [0.4 °C]; median RPE: 11; first and third quartile 11.0–12.0), and natural (38.0 °C [0.3 °C]; median RPE: 10; first and third quartile: 9.5–12.0) grass surface.[12] It is likely that although synthetic grass surfaces transfer more longwave radiation into the surrounding environment, this additional heat load is not enough to cause meaningful changes to core body temperature since the body's thermoregulatory systems are working to avoid such an increase (ie, increased blood flow to the skin to promote cooling). It should also be noted that although the relative humidity was higher over the natural than synthetic grass surface, the absolute humidity was similar (natural: 2.13 Pa vs synthetic: 2.12 Pa). Given that the absolute humidity determines the capacity for sweat evaporation, the similar humidity over both surfaces also likely did not affect heat exchange.

When interpreting the findings from this investigation, the following limitations should be considered. The study was performed in very similar environmental conditions between the 2 days of data collection; however, conditions were not exactly the same. For example, mean differences between day 1 and day 2 of data collection for ambient temperature, relative humidity, and wet-bulb globe temperature were 2.5 °C, 10.3%, and 0.6 °C, respectively. The differences are due to the trials being performed outdoors on separate days, which could not have been performed in more similar environmental conditions, between days. Nevertheless, the minor differences in environmental conditions were accounted for through the use of a counterbalanced study design, and as such, any differences in environmental conditions between days were canceled out through this design. Further, as mentioned previously, athletes were likely seasonally acclimatized, and as such, there may be larger differences in nonacclimatized participants, particularly if they have a lower-whole body sweat rate.

Practical Implications

This study has demonstrated that exercising on synthetic grass surfaces can increase a range of thermoregulatory markers, including skin temperatures, sweat rate, and thermal sensation. However, delta core body temperature and heart rate were not influenced by the surface type; therefore, these findings suggest that exercising on synthetic grass surfaces may not increase thermal or cardiovascular strain in the heat. Further research is required to elucidate the effects across different sports, environments, and populations, as well as implications for performance.

Conclusions

The present study demonstrated that trained/developmental team-sport athletes can experience significantly increased shoulder, chest, and calf skin temperatures, accompanied by a higher sweat rate and ratings of thermal sensation during intermittent exercise on a synthetic compared with a natural grass surface. However, delta core body temperature, heart rate, thermal comfort, and RPE were not significantly different between surfaces. As such, these findings suggest that some markers of heat strain may be increased on synthetic grass, but delta core body temperature, an important outcome measure for thermal strain, was unaffected.

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